$f_\theta(x) = a x_1 + b x_2$

$x_{1,2}$ fixed, a, b uncertain $\quad a \to \bar{a} + \Delta a$

$b \to \bar{b} + \Delta b$

$f_\theta(x)$ a, b correlated

$$(\Delta f_\theta(x))^2 = \sum \frac{\partial f_\theta(x)}{\partial \theta_k} C_{kl} \frac{\partial f_\theta(x)}{\partial \theta_l}$$

C covariance matrix for diagonal = uncorrelated

$C_{kl} = (\Delta \theta_k)^2$

$C_{kl}|_{l \neq k} = 0$

a, b fit together $\to$ correlated

$\Rightarrow$ nearly impossible for high dimensional / large parameter

Why uncertainty

→ needed for Significance/ discovery

→ Likewise needed in ML

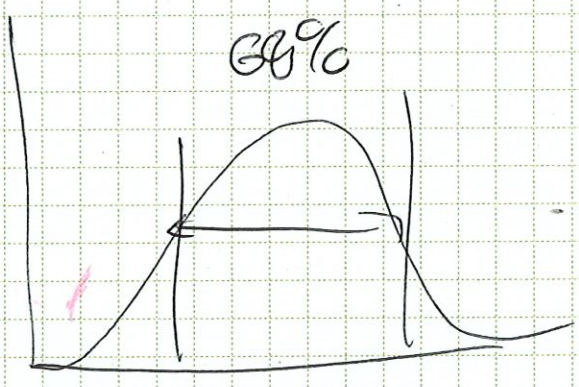
Goal: Field of Uncertainty Quantification

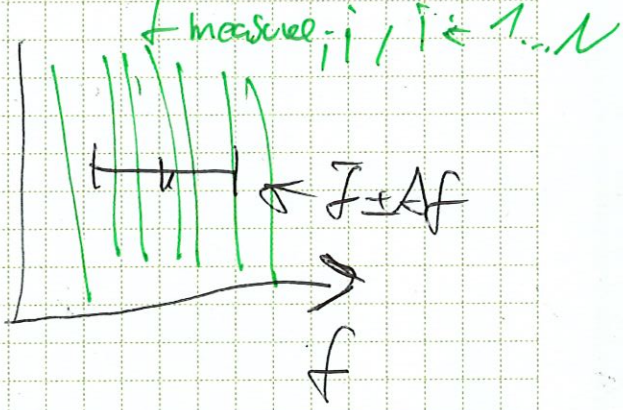
Industry	Physics
trustworthiness of predictions → high unc. → discard	calibration of results/ measurement
wrong uncertainty → Shift fine of ordering holds	wrong uncertainty → false results

Calibrated uncertainty

$f_\epsilon(\delta) = f \pm \Delta f$ ⟵ assumes Gaussian shape

mean ± width





Coverage: $\frac{1}{N} \sum_{i=1}^{N} \mathbb{1}_{f_{measure,i} \in [\bar{f} - \Delta f, \bar{f} + \Delta f]}$

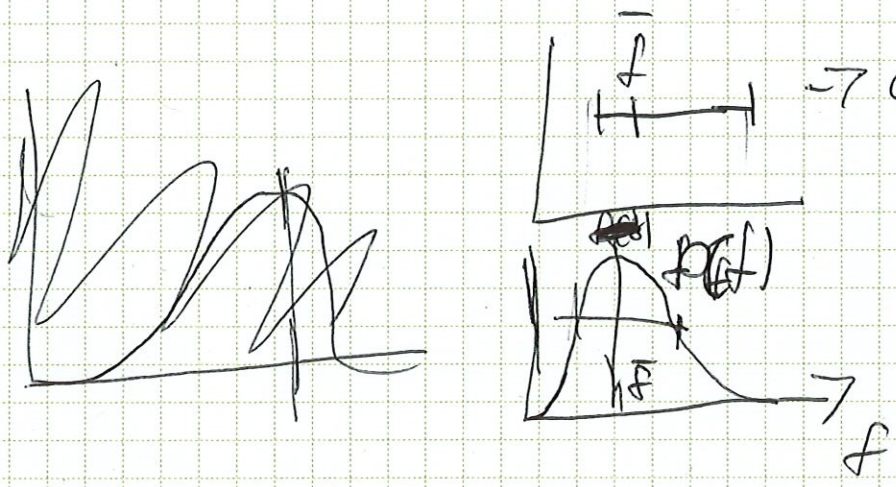
$\mathbb{1}$ indicates function

⇒ for Gaussian Coverage = 68%

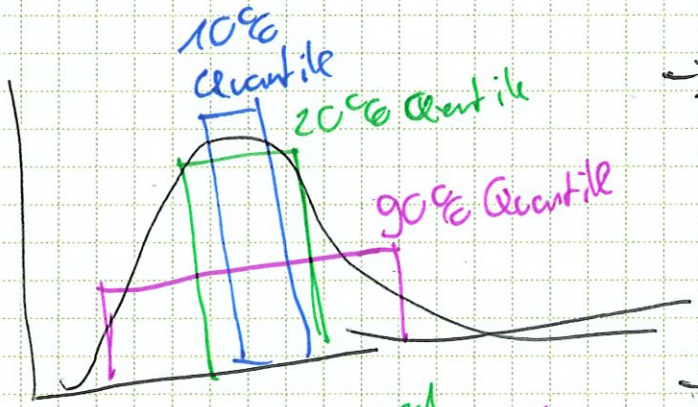
↓

Correct!

Non Gaussian uncertainties?

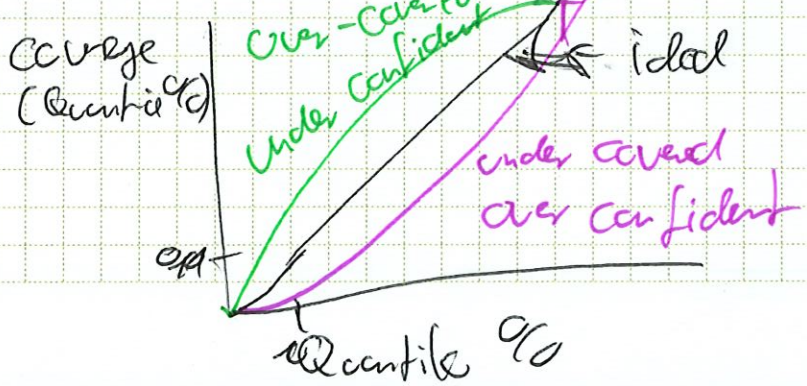


=> calibrate via Quantiles:
Smallest interval containing X % of total probability



→ Don't define symmetric unc. range, define quantiles

→ Coverage (Quantile %)



UQ method 1: Heteroscedastic loss

↗ different uncertainty for different x

Day 1 Gaussian likelihood:

$$-\log p(x_{train}|\theta) = \sum_{x_j \in x_{train}} \left[\frac{(y_j - f_\theta(x_j))^2}{2\sigma_j^2} + \log\left(\frac{1}{\sqrt{2\pi}\sigma_j}\right) \right]$$

↗

$$+ \log \sigma_j + \frac{1}{2}\log(2\pi)$$

Heteroscedastic

$\sigma \to \sigma_\theta(x_j)$

$$L_{Het} = \frac{1}{N} \sum_j \left[\frac{(y_i - f_\theta(x_i))^2}{2\sigma_\theta(x_j)} + \log \sigma_\theta(x_j) \right]$$

↗ mean of the batch

↖ potential divergence $\sigma_\theta \to \infty$, $L \to 0$

↖ regularizes

$NN_\theta(x_i) \nearrow f_\theta(x_i)$

$\searrow \sigma_\theta(x_i)$

⇒ network learns mean and uncertainty

want $\rho > 0$ → don't learn ρ,

learn S $\rho = \exp(S)$

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Method 2: Bayesian Neural Networks

Training: want to predict label of new point

$x \notin x_{train}$, using x_{train}, y_{train}

$$p(y|x, x_{train}) = \int d\theta \; p(y, \theta | x, x_{train})$$

$$p(y, \theta | x, x_{train}) = p(y | x, \theta, x_{train}) \cdot p(\theta | x, x_{train})$$

→ computing likelihood

$$\int d\theta \; p(y | x, \theta, x_{train}) \; p(\theta | x, x_{train})$$

↑ no training set during eval

↖ no eval point in training

Likelihood of y for a given model

Likelihood of a certain model given training data

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reform Arg Dep?

$$p(\theta|x_{train}) = \frac{p(x_{train}|\theta)p(\theta)}{p(x_{train})}$$

$$p(x_{train}) = \int d\theta\, p(x_{train}|\theta)p(\theta)$$

=> last time -> argmax -> 1 set of θ

Now: need θ distribution

=> don't know, approximate

$$p(\theta|x_{train}) \approx q(\theta)$$

$$p(y|x, x_{train}) = \int d\theta\, p(y|x,\theta)\, q(\theta)$$

$q(\theta)$: common choices, Gaussian parameter, each parameter independent

$$q(\theta) = \prod_k \mathcal{N}(\theta_k|\mu_k,\sigma_k)$$

each parameter defined by mean and width

D_{KL}: KL divergence:

$$D_{KL}[q(x), p(x)] = \int dx\, q(x) \log\left(\frac{q(x)}{p(x)}\right)$$

=> loss for matching ~~fits~~ ~~distributions~~ predictions

$$D_{KL}[q(\theta), p(\theta|x_{train})] = \int d\theta\, q(\theta) \log \frac{q(\theta)p(x_{train})}{p(\theta)p(x_{train}|\theta)}$$

$$= \int d\theta\, q(\theta) \log \frac{q(\theta)}{p(\theta)} - \int d\theta\, q(\theta) \log(p(x_{train}|\theta)) + \log p(x_{train}) \int d\theta\, q(\theta)$$

$$= D_{KL}[q(\theta), p(\theta)] - \int d\theta\, q(\theta) \log(x_{train}|\theta) + \log p(x_{train})$$

↗ discard all that does not depend on θ

Loss

$$= -\int d\theta\, q\theta \log p(x_{train}|\theta) + D_{KL}[q(\theta), p(\theta)]$$

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$$= -\int d\theta\, q(\theta) \log p(x_{train} | \theta) + D_{KL}[q(\theta), p(\theta)]$$

$$= \sum_i \int d\theta\, q(\theta) \log p(y_i | x_i, \theta) + \text{const}$$

↓ Gaussian likelihood

$$= \sum \left\langle \frac{|y_i - f_\theta(x_i)|^2}{2\sigma^2} \right\rangle_{q(\theta)}$$

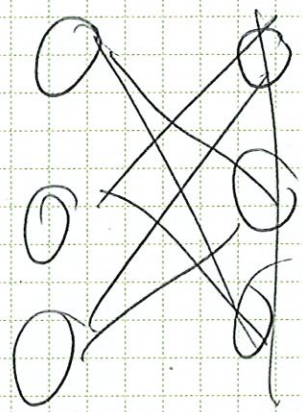
↖ numerically: sample $\theta_i \sim q(\theta)$, repeat

$p(\theta)$: prior to physics info → hyperparameter

→ $\mathcal{N}(0,1)$

→ KLD pulls $q(\theta) \to \mathcal{N}(0,1)$ → regularization

$w_{ij} \sim \mathcal{N}(\mu_{ij}, \sigma_{ij})$



→ $f_\theta(x) \Big|_{\theta \sim q(\theta)}$ → one Sample

$\bar{f}_\theta$ $\langle f_\theta(x) \rangle_{q(\theta)}$ → many samples of height, same Pixel

→ mean Prediction

Δf_θ^2 $\mathrm{Var}_{q(\theta)}(f_\theta(x))$

→ mean Uncertainty

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